Capstone Project Weekly Journal

Three Degree of Freedom Stewart Platform Control and Visualization

Student: Vu Tran Khoi Nguyen
Student ID: s3978262
Project Team: The Boys
Journal Period: Weeks 1–12 | 1 July–20 September 2026
Project Partner: Bosch Rexroth

This journal records my weekly technical development, individual contributions and the main issues encountered while developing the Bosch ctrlX CORE Virtual, PLC, HMI and 3D Viewer subsystems, together with the integration and verification of the Stewart Platform project.

Week 1

Meeting date: 1 July 2026
Journal period: 1–7 July 2026

Key project activities

- The team reviewed the work completed during the semester break and created a completion plan for the remaining project period.
- We confirmed that the platform would demonstrate three degrees of freedom: heave, roll and pitch.
- The project was divided into several workstreams, including kinematics and simulation, physical actuator control, Bosch ctrlX programming and visualization, mechanical construction and system integration.

Key individual contributions and accomplishments

- I helped review the overall project requirements and the proposed system architecture.
- I began identifying the role of the Bosch ctrlX CORE within the planned control and visualization system.
- I reviewed the relationship between the user inputs, inverse kinematics, actuator commands and the virtual platform model.

Major issues

- The definition of the digital twin and its expected communication with the physical system was not yet clear.
- The final hardware configuration was still being decided.

- Several parts of the project depended on hardware that was not yet available.

Week 2

Meeting date: 8 July 2026
Journal period: 8–14 July 2026

Key project activities

- The team refined the completion plan and system architecture.
- Hardware requirements and their purposes were reviewed.
- The team started studying the Bosch manuals and the virtual ctrlX environment.

Key individual contributions and accomplishments

- I started studying the Bosch Rexroth documentation and ctrlX CORE environment.
- I investigated how PLC variables could be used to control the Stewart Platform.
- I helped consider how the HMI and virtual model could fit into the overall system.

Major issues

- The roles of the physical ctrlX CORE, virtual ctrlX CORE and STM32 were not yet clearly separated.
- The communication method between the different systems was still uncertain.
- The team needed to continue development while waiting for hardware availability.

Week 3

Meeting date: 15 July 2026
Journal period: 15–21 July 2026

Key project activities

- The team reviewed the hardware requirements and intended application of the Stewart Platform.
- More research was conducted into Stewart-platform kinematics.

- The team began developing ideas for both the working prototype and industrial system.

Key individual contributions and accomplishments

- I contributed to the initial working-prototype concept together with Khang and Quan.
- I continued investigating Bosch ctrlX CORE PLC programming.
- I studied how the Stewart Platform's heave, roll and pitch inputs could be represented in the virtual control system.
- I helped understand how the virtual model could later be connected to the PLC variables.

Major issues

- The project still had uncertainty about the final hardware configuration.
- The Stewart Platform kinematics were relatively complex and required further understanding before the virtual model could be implemented correctly.
- The lack of physical Bosch hardware was becoming an important project risk.

Week 4

Meeting date: 22 July 2026
Journal period: 22–28 July 2026

Key project activities

- The team presented the working prototype and industrial-system concepts.
- We confirmed that the physical Bosch ctrlX CORE was not available.
- Alternative approaches were discussed to continue the project without the physical Bosch controller.

Key individual contributions and accomplishments

- I helped investigate alternatives for the missing physical ctrlX CORE.
- I continued working on the virtual ctrlX CORE approach.
- I contributed to the working-prototype development and helped consider how it could be used to verify the inverse-kinematics behaviour.
- I helped identify what could still be demonstrated using the virtual Bosch environment.

Major issues

- The missing physical ctrlX CORE prevented the original HIL implementation.
- The team had to separate the physical STM32 development from the virtual Bosch development.
- This created an additional future communication and synchronization requirement.

Week 5

Meeting date: 29 July 2026
Journal period: 29 July–4 August 2026

Key project activities

- The team reviewed inverse-kinematics calculations and the mechanical dimensions of the platform.
- The Arduino prototype and rack-and-pinion mechanism were tested.
- Physical motor-control development started moving from concept toward practical testing.

Key individual contributions and accomplishments

- I continued working on the virtual control concept and the Bosch environment.
- I reviewed how the actuator lengths calculated by the inverse kinematics could be represented in the virtual platform.
- I participated in discussions about the platform dimensions and how these dimensions affected the virtual model.
- I continued studying PLC programming and the requirements for controlling the platform in the 3D Viewer.

Major issues

- The mechanical dimensions were still being refined.
- The virtual model had to remain consistent with the changing physical design.
- The lack of a physical ctrlX controller continued to limit the planned HIL system.

Week 6

Meeting date: 5 August 2026
Journal period: 5–11 August 2026

Key project activities

- The team reviewed the working prototype and inverse-kinematics results.
- Industrial platform dimensions and actuator requirements were refined.
- The Bosch 3D Viewer and ctrlX CORE Virtual development were discussed together with the alternative HIL approach.

Key individual contributions and accomplishments

- I continued developing the Bosch virtual-system approach.
- I worked with the team to understand how the platform geometry and actuator requirements should be represented in the 3D Viewer.
- I contributed to discussions about controller alternatives and the role of the STM32 in replacing the unavailable physical ctrlX CORE.
- I also helped investigate how the Bosch virtual system could eventually communicate with the physical system.

Major issues

- The physical ctrlX CORE was still unavailable.
- The project now had two parallel control paths: the physical STM32 system and the virtual Bosch system.
- A reliable communication method would be required to combine them.

Week 7

Meeting date: 12 August 2026
Journal period: 12–18 August 2026

Key project activities

- The team reviewed the Bosch 3D Viewer model and implementation of inverse kinematics.
- The STM32 was confirmed as the controller for the physical actuator demonstration.
- Work continued on PLC variables, platform inputs and the virtual model.

Key individual contributions and accomplishments

- I continued developing the virtual ctrlX CORE and 3D Viewer side.
- I worked with Khang and Quan on the Stewart Platform model in the Bosch 3D Viewer.

- I helped investigate how inverse-kinematics outputs could be represented as PLC variables for the virtual platform.
- I also worked with Dung on investigating communication between the STM32 and Simulink.

Major issues

- The virtual and physical systems were becoming increasingly independent because of the missing physical Bosch hardware.
- The communication interface between STM32 and the simulation environment still needed to be established.
- The PLC variables and 3D Viewer transformations had to be carefully matched to the Stewart Platform geometry.

Week 8

Meeting date: 19 August 2026
Journal period: 19–25 August 2026

Key project activities

- The team reviewed the STM32 inverse-kinematics implementation and planned the PIL testing process.
- Communication between the STM32 and Simulink was discussed.
- The Bosch PLC and 3D Viewer development continued in parallel.

Key individual contributions and accomplishments

- I worked with Dung and Toan on establishing the STM32–Simulink communication approach.
- I contributed to the planning of PIL testing and how the calculated actuator strokes could be compared with the Simscape model.
- I continued the PLC and 3D Viewer implementation and helped test the virtual platform using different input values.
- I reviewed the overall architecture to make sure the virtual and physical paths were clearly defined.

Major issues

- Communication between the different systems remained one of the main integration challenges.

- The schedule limited the time available to implement and test a complete communication protocol.
- The team had to prioritize stable subsystem demonstrations.

Week 9

Meeting date: 26 August 2026
Journal period: 26 August–1 September 2026

Key project activities

- The physical STM32 actuator-control system was tested alongside the Bosch PLC and 3D Viewer.
- The team investigated possible Ethernet communication between the STM32 and ctrlX CORE.
- The physical and virtual subsystems were tested independently.

Key individual contributions and accomplishments

- I continued the PLC and 3D Viewer implementation with Khang and Quan.
- I tested the virtual Stewart Platform using different heave, roll and pitch inputs.
- I contributed to the initial PIL testing with Toan and the other team members.
- I also investigated the remaining communication issues between the physical and virtual systems.

Major issues

- Ethernet and communication setup remained difficult to complete within the available time.
- The virtual ctrlX system and physical STM32 system were still not synchronized.
- Testing had to be performed separately to isolate problems in each subsystem.

Week 10

Meeting date: 2 September 2026
Journal period: 2–8 September 2026

Key project activities

- The physical actuator and STM32 control system were tested further.

- The Bosch 3D Viewer was tested using different heave, roll and pitch inputs.
- The team began preparing the final report and presentation evidence.

Key individual contributions and accomplishments

- I tested the PLC program and 3D Viewer movement using different input values.
- I checked that the virtual platform responded correctly to valid heave, roll and pitch commands.
- I helped review the remaining difference between the virtual system and the physical actuator system.
- I contributed to documenting the system architecture and limitations for the final report.

Major issues

- The physical and virtual systems still operated independently.
- A complete end-to-end digital twin could not be demonstrated.
- The remaining time had to be divided between technical testing, documentation and presentation preparation.

Week 11

Meeting date: 9 September 2026
Journal period: 9–15 September 2026

Key project activities

- The final physical actuator and virtual 3D Viewer results were reviewed.
- The team confirmed that the two systems were functioning independently.
- Work shifted towards final documentation, presentation preparation and demonstration planning.

Key individual contributions and accomplishments

- I helped finalize the virtual ctrlX CORE and 3D Viewer demonstration.
- I reviewed the final virtual-system results and helped ensure that the project did not claim a completed real-time digital twin when the physical and virtual systems were not connected.
- I contributed to the final report, presentation material and explanation of the integration limitations.

- I also helped prepare the virtual system for the final demonstration.

Major issues

- There was insufficient time to complete and validate the STM32-to-ctrlX communication.
- The project therefore had to clearly distinguish between the functioning physical subsystem and the functioning virtual subsystem.
- Evidence for the final presentation needed to clearly explain what had and had not been integrated.

Week 12

Meeting date: 16 September 2026
Journal period: 16–20 September 2026

Key project activities

- The team confirmed the final outcomes: the virtual ctrlX CORE successfully controlled the Stewart Platform visualization in Bosch 3D Viewer, while the STM32 successfully controlled the physical linear actuator.
- The final report and presentation were completed.
- The team rehearsed the presentation and prepared the final demonstration and exhibition.

Key individual contributions and accomplishments

- I finalized and tested the virtual ctrlX CORE and 3D Viewer demonstration.
- I contributed to the final presentation and technical explanation of the Bosch virtual system, HMI, integration limitations and future work.
- I reviewed the final report and helped ensure that the terminology accurately represented the achieved system.
- I helped prepare the virtual demonstration and backup evidence for the exhibition.

Major issues

- The physical and virtual systems remained independent at the end of the project.
- The 3D Viewer should therefore be described as a PLC-controlled virtual model/open-loop visualisation rather than a fully synchronized digital twin.

- Future development should focus on connecting the STM32 to ctrlX CORE, completing three-actuator synchronization and adding physical feedback for proper digital-twin validation.